

An Evidence-Driven ePolyScat Research Portal

End-to-End Scientific Workflows, Reproducible Inputs, and Verified Computation

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PROJECT ACCEPTANCE
REVIEW
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Scientific Gateway Delivery

RESEARCH GAP

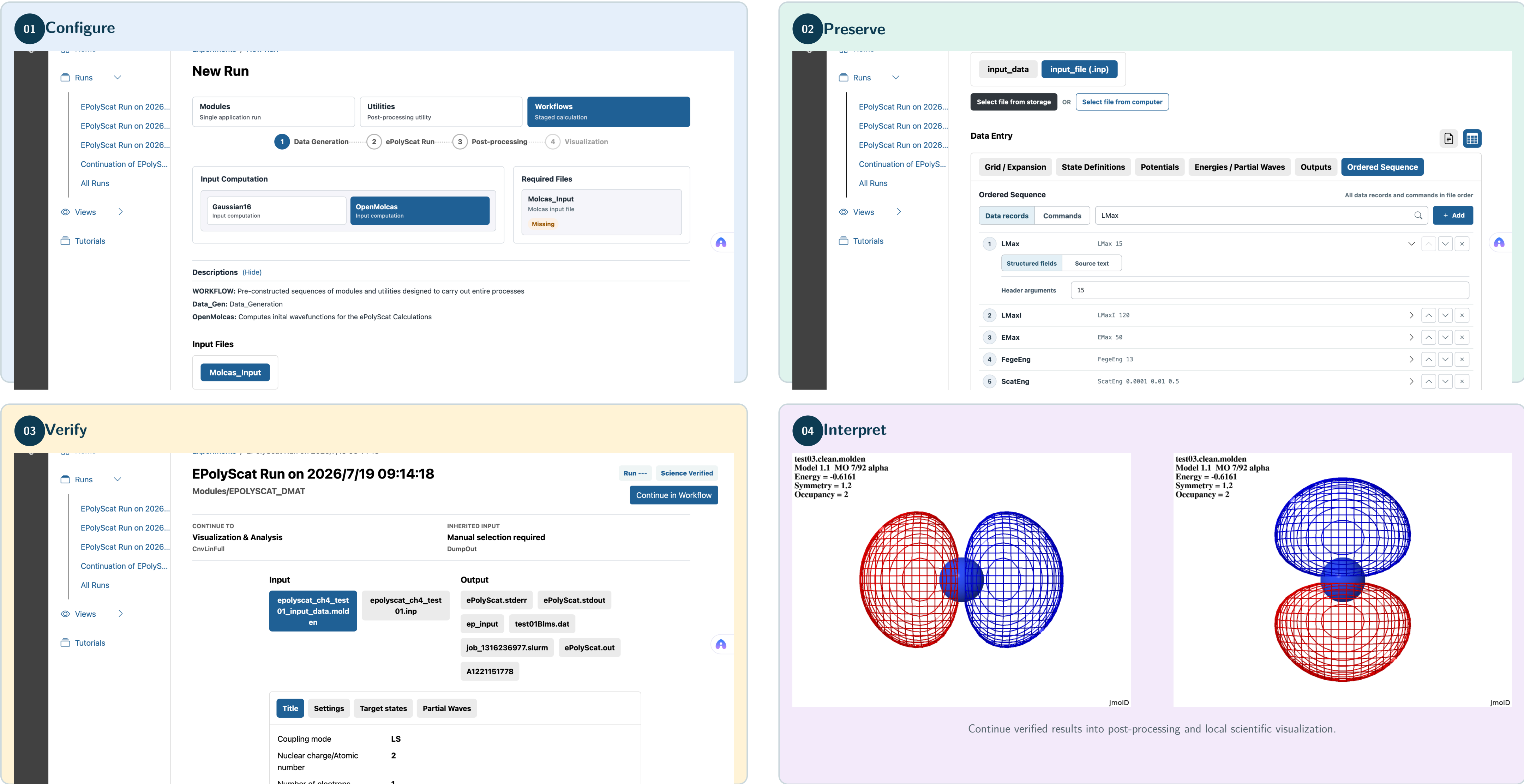
Scientific workflows span specialized programs, remote schedulers, and local analysis tools. Job completion alone does not prove that scientifically valid outputs were produced or that the correct files reached the next stage.

CONTRIBUTION A · ZIMING ZHANG Workflow Portal and Scientific Verification

KEY CONTRIBUTION

We transformed ePolyScat from a collection of disconnected tools into a guided, inspectable workflow. Each stage preserves the researcher's original input, records output provenance, and exposes a verified handoff to the next calculation or visualization step.

Evidence narrative: from input to interpretation



Separate scheduler completion from scientific success using application-specific output evidence.

Engineering decisions

- Lossless ordered document model for reproducible ePolyScat inputs.
- Semantic output roles instead of filename-only inference.
- Resource-aware execution contracts across supported HPC systems.
- Provenance-aware continuation with researcher review before submission.

Scientific value

Researchers can reproduce exactly what was submitted, distinguish scheduler artifacts from scientific results, and continue a calculation without manually rebuilding file dependencies.

Reproducible input → interpretable output

ACCEPTANCE CLAIM

Connect configuration, execution, scientific validation, and stage-to-stage handoff in one portal experience.

CONTRIBUTION B · NGUYEN, THINH M [Partner Progress Title]

[One sentence: what changed, for whom, and why it matters.]

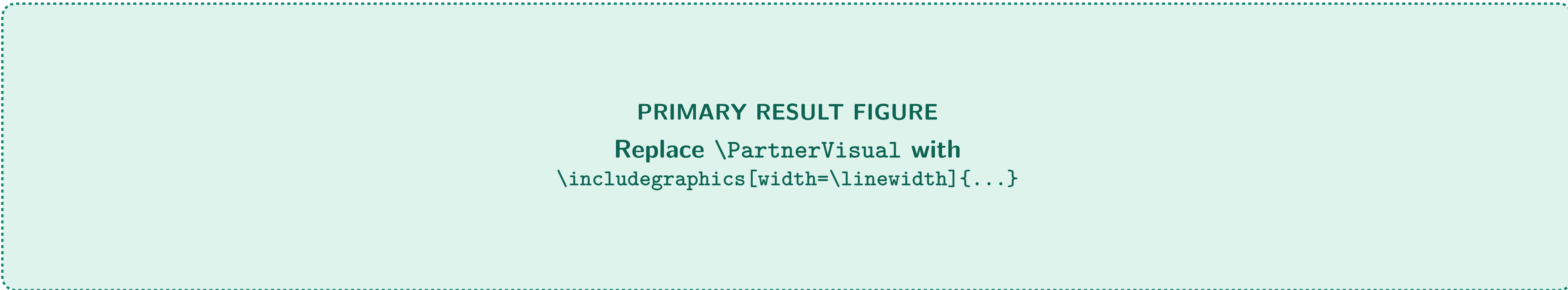
Problem & method

[Problem and method: summarize the approach in two or three concise points.]

Implementation

[Implementation or experimental contribution: name the system, artifact, or analysis delivered.]

Primary result



[One sentence explaining what the visual proves.]

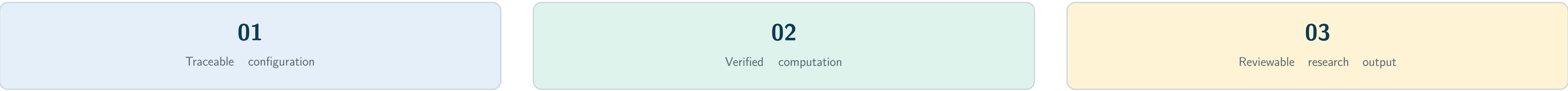
Validation evidence

✓ [Tests, measurements, comparison, or user evidence supporting the claim.]

Outcome metrics

[Two to four defensible outcome metrics, separated by centered dots.]

How this contribution fits the shared objective



REPLACEMENT GUIDE Edit the eight \Partner... macros near the top; every labeled area updates automatically.